MSB7103: Bio-Unix & Shell Scripting (4 CU)

Course Description:

Unix is an operating system commonly used in bioinformatics analysis and programming. Many tools used in bioinformatics work best when run in a Unix-based, command line environment. As such, working in a Unix environment is a fundamental skill needed by all bioinformatics and genomics researchers especially the ability to work productively in a Unix based, command-line environment. This course will provide students with a working knowledge of computer systems, programming and interfacing with other software essential to a career in bioinformatics.

Course Objectives:

At the end of this course, the student should be able to:

1. Explain the Unix Shell and command-line environment
2. Outline procedures of obtaining and installing command-line utilities
3. Discuss data processing and visualization techniques
4. Perform shell scripting and develop pipelines or workflows using Shell scripting

Course Learning Outcomes:

By the end of this course, students should be able to:

1. Comfortably operate within the Unix Shell and command-line environment 2. Identify and appropriately obtain, install, utilize and troubleshoot command-line utilities 3. Develop analytical pipelines and workflows using Shell scripting
4. Process data sets in a pipeline of linked steps and checking results using a visualization tool

Indicative Content:

No.	Content	Contact Hours
1.	Introduction to Linux/Unix, commands and the file system	10
2.	Introduction to UNIX editors	10
3.	Programming Utilities	10
4.	Installing tools	5
5.	Manipulating files and Data on Linux	5
6.	Shell Scripting/Programming	10
7.	Using remote machines (servers, clusters, clouds)	10
	TOTAL	60

Mode of Delivery:

This course will be delivered as a participatory, hands-on practical sessions, and active learning. Learning will be facilitated through lectures, tutorial videos, Individual and group work comprising of assignments and prescribed reading.

Mode of Assessment:

1. Group Assignments 10%
2. Individual Assignments 20%
3. Course Assessment Tests 10%
4. Final examination 60%

Reading List/References:

1. Learning the bash Shell, (1998), 2nd Edition - O'Reilly Press
2. UNIX for Dummies Quick Reference, (1998), 4th Edition by Margaret Levine Young, John R. Levine
3. Computational Biology, (2004): Unix/Linux, Data Processing and Programming by Röbbe Wünschiers